

Revenue Cloud

Product Configurator

Enablement Exercises

Version 1.0, Winter '26



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Release Overview

In this release enablement exercise we want to highlight the **CML Guide** - Learn Constraint Modeling Language (CML) Core Concepts, Best Practices, and More

Salesforce Revenue Cloud Feature Name includes the following new features in the release:

1. **Admin UI- Constraint Template Enhancements** - Improve Constraint Definition with Flexible Expression-Building Capabilities in Visual Builder
2. **Support Product Component Groups in CML** - Maintain Data Integrity in Configuration Rules by Importing Product Component Groups to Constraint Builder



3. **Visual Builder- new rule templates** - Use New Rules in Visual Builder for More Configuration Options
4. **Dynamic Loader/Table Constraints** - Leverage Data from a Salesforce Object to Use In a Constraint Model
5. **Non-Blocking UI** - Enable Non-Blocking Behavior in the Configurator
6. **Rule based Product Recommendation** - Add Product Recommendations During Quote and Order
7. **Product Centric Modelling** - Keep Constraint Model CML in Sync with the Latest Product Definitions



Pre-requisites

To initiate the process of creating constraints and rules for your product configuration, the first step involves importing the product or configurable bundle, along with its components and attributes, into the Visual Constraint Builder. This ensures that the Visual Constraint Builder has complete information of the product structure and attributes.

For the purpose of this exercise, we will import the "LaptopProBundle" product from the product catalog into the Visual Constraint Builder environment.

Design Time Configuration

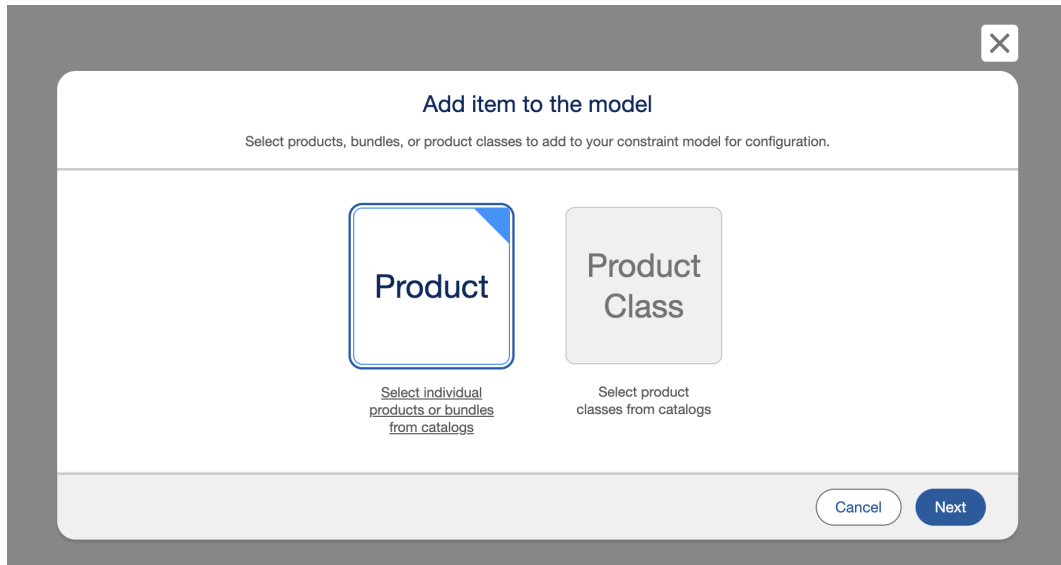
1. In the Product Catalog Management home page, scroll down to the 'Rules' Section and click in the tile 'Constraint Models'
2. On the Constraints home page click the 'New Constraint Model' button
3. In the modal enter the following information:
 - a. Name: 'ComputerModel'
 - b. Usage Type: 'Constraint'
 - c. Context definition: 'PricingTransactionCD2'
 - i. *The context definition that is used in the RCA default pricing procedure from Revenue Settings should be the one used here.
4. Click Save.
5. A new constraint model with version 1 called "ComputerModel V1" will be created.
6. Click on the constraint model Version Name (ComputerModel V1) to launch the Visual Builder.
7. In the Visual Builder, Click on "Add Item" in the left panel.

The screenshot displays the Visual Builder interface. At the top, there is a dark blue header bar with a 'Constraint Builder' icon and label on the left, and 'DemoModel V1' on the right. Below this, a 'Visual Builder' button is visible. The main content area features a 'Product Index' panel on the left, which includes a search bar labeled 'Search this list...' and a prominent 'Add Item' button. The 'Add Item' button is highlighted with a blue border and a shadow effect.

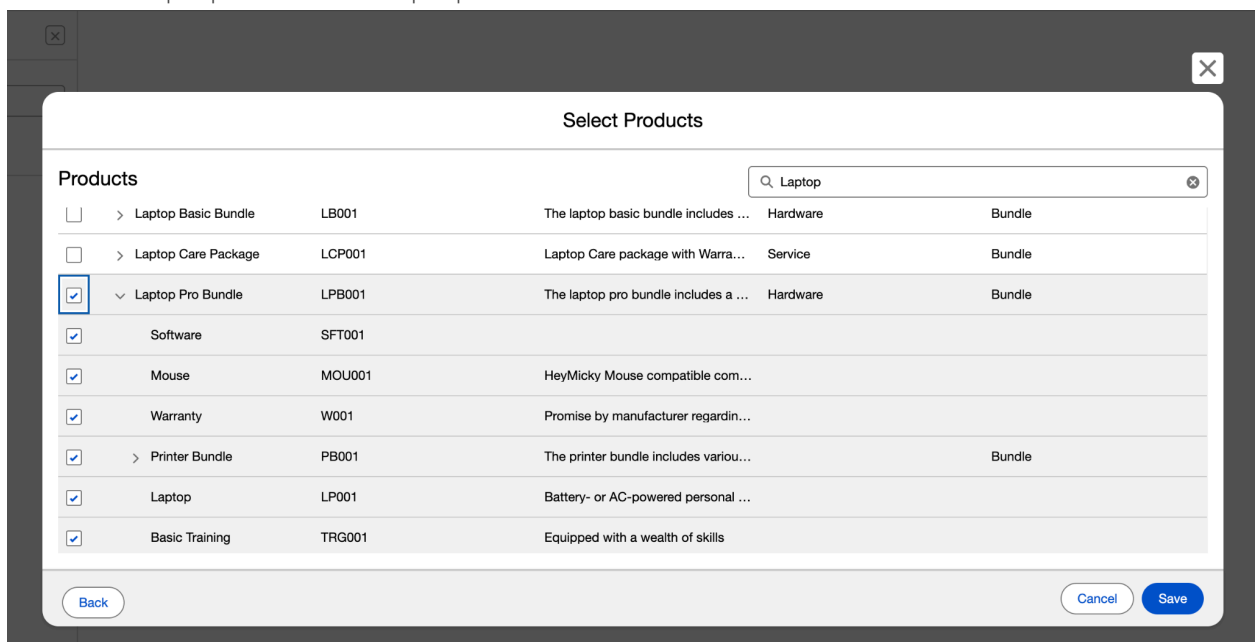


8. In the resulting modal select “Product” and click ‘Next’.

**Note: Product Class option is currently not available.*



9. Search for laptop and Select Laptop Pro Bundle from the results



10. Click Save.



Laptop Pro Bundle should be imported into the Visual Builder and you can view the bundle and all components on the left panel

Constraint Builder

model1 V1

Visual Builder

*

Product Index

×

Q Search this list...

Add Item

Products

⌵

⌵ LaptopProBundle

>

↗ SoftwareGroup

☑

↗ AccessoriesGroup

↔

Mouse

↔

Warranty

>

↔ PrinterBundle

>

↗ LaptopGroup

◦

Software

◦

Mouse

◦

Warranty

>

⌵ PrinterBundle

◦

Printer

◦

PrinterPaper

◦

Laptop

◦

BasicTraining



Reference CML Guide

Build your skills with CML, the language used with Constraint Rules Engine to define constraint models. The [CML User Guide](#), available as a PDF, covers key CML concepts like working with types, variables, and relationships, and using constraints to enforce rules and conditions. The guide includes code samples, best practices, and debugging tips, and is updated regularly with the latest CML features and capabilities.

Dynamic Loader/Table Constraints

Business Objective

Import data from a standard or custom Salesforce object to use as a table constraint in a constraint model. The imported data populates the columns and rows in the table constraint in CML, and saves the step of manually entering the data.

Use Cases

Example Use Cases

Use Case 1:

The product offering focuses on specific different combinations of valid attributes values that can be a choice for configuration. For example, a laptop can only be sold with RAM 1GB, HD 500GB and Intel graphics, or RAM 3 GB, HD 1TB and Intel graphics II. In this example there are only 2 valid combinations, however with Table constraint dynamic loader, there could be potentially many valid combinations.

Design Time Configuration

1. Create a custom object via setup.
 - a. Custom object Name: DLTable.
 - b. Plural Name DLTables
 - c. Select "Launch New Custom Tab Wizard after saving this custom object" checkbox



- d. Click Save
 - e. Click Next
 - f. Choose a tab style and click Next, Next and Save
2. Add text field (length 100) and name it Display
3. Add another text field (length 100) and name it Display_size
4. Navigate back to setup home and search for permission sets in the quick find search box
5. Navigate to the Permission sets page and locate Advanced Configurator Designer Licenseless permission set
6. Click on it and then click on Object Settings.
7. Locate DLTables and click on it
8. Enable all the permission available:
 - a. Create
 - b. Read
 - c. Update
 - d. Delete
 - e. View All Records, View All Fields, Modify All

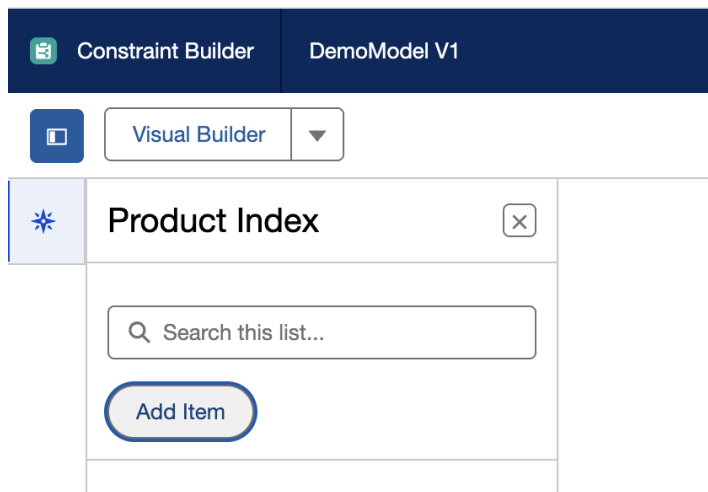
Object Permissions

Permission Name	Enabled
Read	☒
Create	☒
Edit	☒
Delete	☒
View All Records	☒
Modify All Records	☒
View All Fields	☒

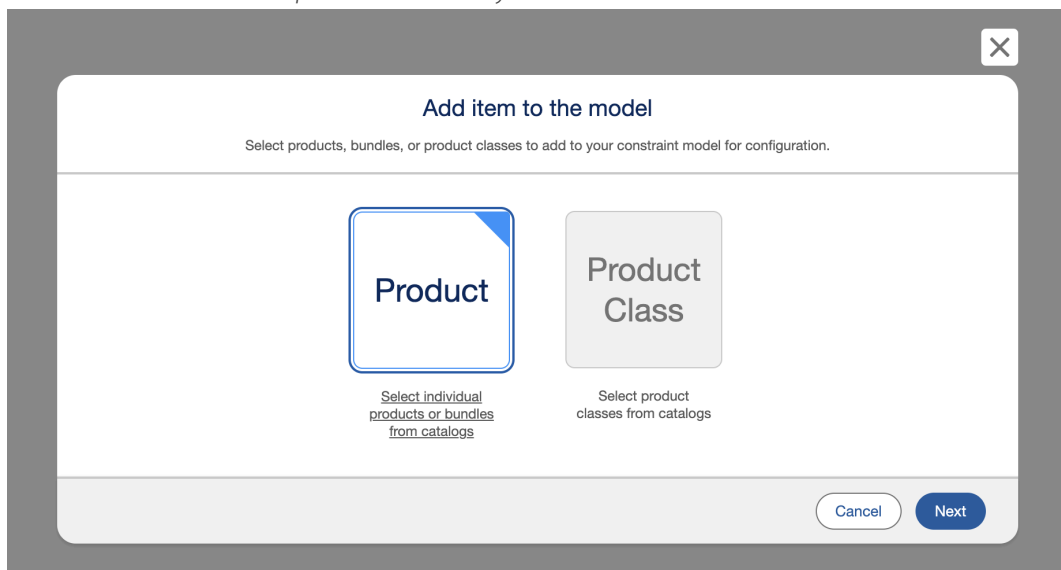
9. Save the permissions.
10. Create DLTable records with Display and Display_size by launching the DLTable via the App Launcher
 - a. 2k Built-in Display and 24 Inch
 - b. 4k Built-in Display and 27 Inch
11. In the Product Catalog Management home page, scroll down to the 'Rules' Section and click in the tile 'Constraint Models'
12. On the Constraints home page click the 'New Constraint' button
13. In the modal enter the following information:
 - a. Name: 'ComputerModel'



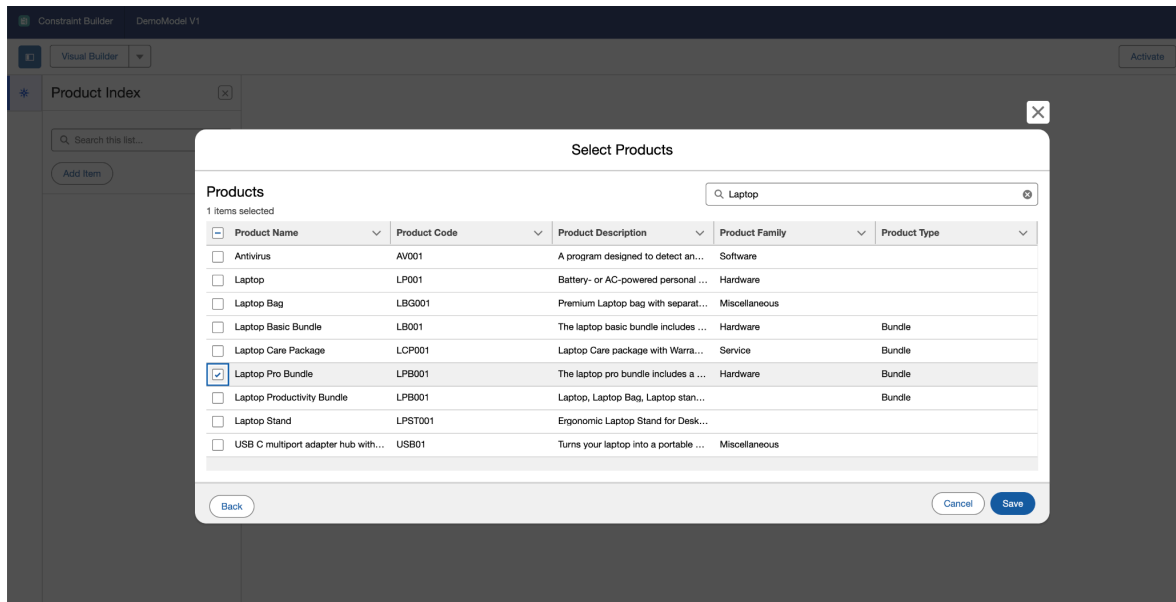
- b. Usage Type: 'Constraint'
 - c. Context definition: 'PricingTransactionCD1'
 - i. *The context definition that is used in the RCA default pricing procedure from Revenue Settings should be the one used here.
14. Click Save.
15. A new constraint model with version 1 called "ComputerModel V1" will be created.
16. Click on the constraint model Version Name (ComputerModel V1) to launch the Visual Builder.
17. In the Visual Builder, Click on "Add Item" in the left panel.



18. In the resulting modal select "Product" and click 'Next'.
- *Note: Product Class option is currently not available.*

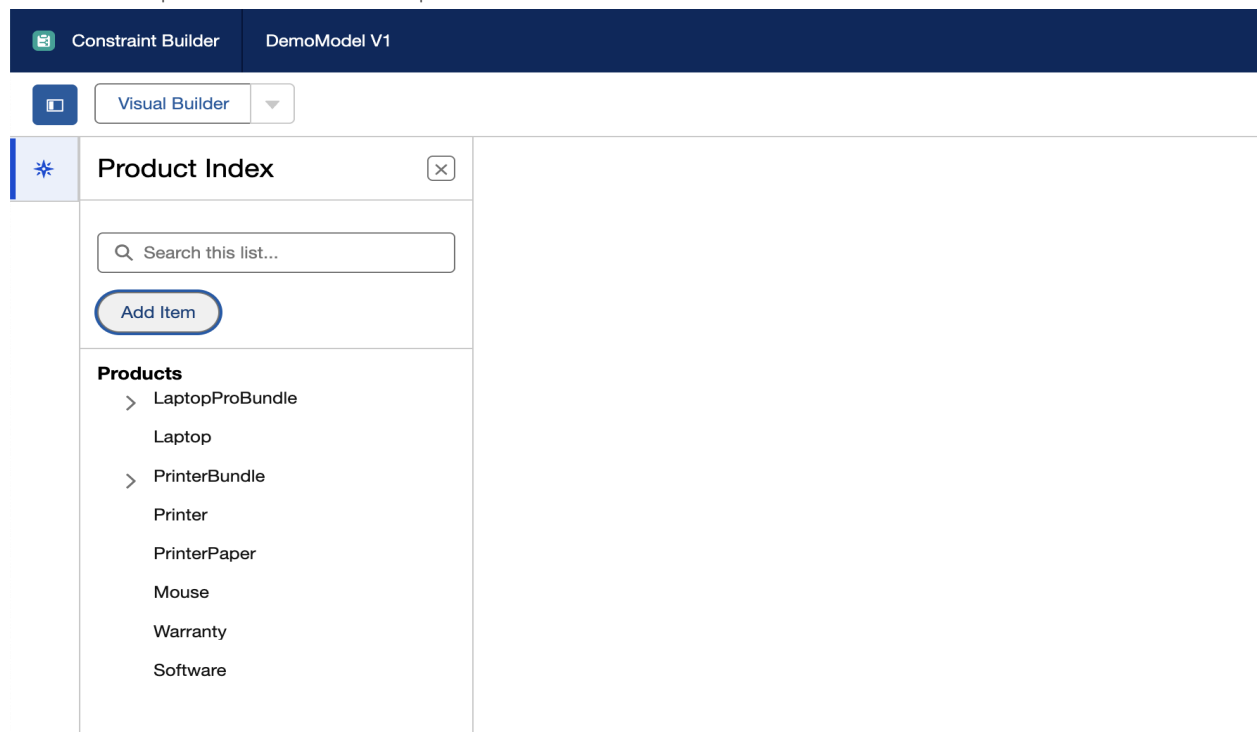


19. Search for laptop and Select Laptop Pro Bundle from the results



20. Click Save.

Laptop Pro Bundle should be imported into the Visual Builder and you can view the bundle and all components on the left panel



21. Click on the arrow to the side of the visual builder and switch to CML Editor

22. Within the Laptop type definition section add (copy and paste):



```
constraint(table(Display, Display_Size,  
SalesforceTable("DLTable__c", "Display__c,Display_Size__c")), "Display and Display_Size  
combination is incorrect");
```

Syntax:

```
constraint(table(attr1, attr2, attr3, attr4,  
SalesforceTable("tablename__c", "column1name__c, column2name__c, co  
lumn3name__c, column4name__c")));
```

23. Save and Activate

Preference Rule with Expression Grouping

Business Objective

Configuration preferences can be set based on predefined conditions. These conditions can be overridden or adjusted by users to align with their specific requirements.

Use Cases

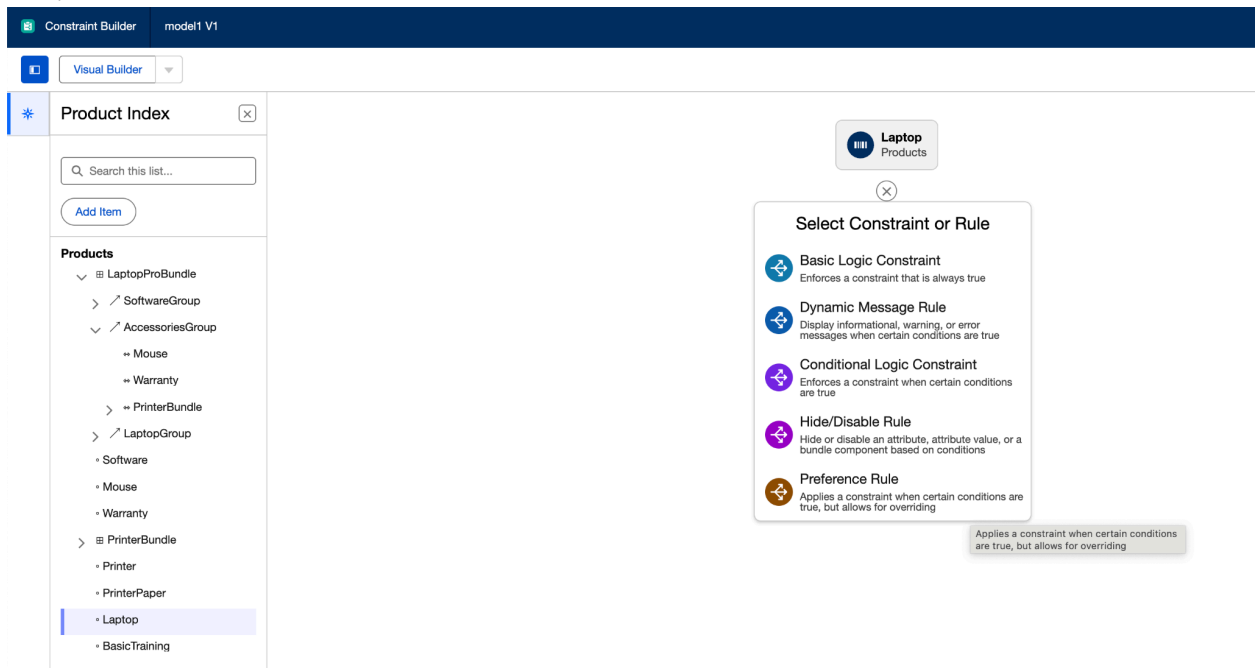
This example uses the bundle product LaptopProBundle, which includes a Laptop component. The preference rule automatically sets the Display resolution to '4k Built-in Display' when the screen size is '27 Inch' during configuration runtime, if possible. The key distinction between a preference rule and a constraint is in the ability of sales representatives to override the value set by a preference rule. Once the preference is overridden by the user, the system learns that and wouldn't run that preference rule for that quote line anymore.

Design Time Configuration

1. Open the LaptopProBundle constraint model in the Visual Builder.
2. Check if the model is Active and Deactivate it before proceeding to edit.
3. In the Visual Builder Left panel, select Laptop under the list of Products



4. In the main panel, click the '+' icon under the Laptop to view the available constraint templates.



5. Select Preference Rule.
6. Write a constraint Summary : “Set 4K Display preference”
7. In the top section of the constraint Card click ‘+ Add Expression’
8. Complete the expression definition as follows:
 - a. **LHS Element:** Laptop : Attribute: Windows Processor
 - b. **Operator:** == Equals
 - c. **RHS Element:** Value: Intel Core i9 5.2 GHz
9. Repeat steps 7 and 8 as follows:
 - a. **LHS Element:** Laptop : Attribute: Graphics
 - b. **Operator:** == Equals
 - c. **RHS Element:** Value: MSI Gaming GeForce RTX 3060
10. Repeat steps 7 and 8 as follows:
 - a. **LHS Element:** Laptop : Attribute: Display_Size
 - b. **Operator:** == Equals
 - c. **RHS Element:** Value: 27 Inch
11. Edit the grouping logic as follows :
 - a. (1 or 2) and 3



12. In the bottom section (after prefers) of the constraint Card 'Add Expression' Complete the expression definition as follows:

- Variable Type: Attribute
- Variable: Display
- Operator: == Equals
- Value: 4k Built-in Display

13. Enter a run-time Message: 4K Display Preferred.

The screenshot shows the Salesforce Constraint Editor interface. On the left is a sidebar with a 'Products' section containing a tree view: LaptopProBundle, Software, Mouse, Warranty, PrinterBundle, Printer, PrinterPaper, Laptop (highlighted), and BasicTraining. The main editor area is titled 'Set 4K Display preference'. It contains a 'Preference Rule' section with a description: 'Applies a constraint when certain conditions are true, but allows for overriding.' Below this are three expression rows, each with an 'Element', an 'Operator' (set to '== Equals'), and a 'Value'. Row 1: Element 'Laptop > Windows_Processor', Operator '== Equals', Value 'Select...'. Row 2: Element 'Laptop > Graphics', Operator '== Equals', Value 'Value > MSI Gaming GeForce RTX 3060'. Row 3: Element 'Laptop > Display_Size', Operator '== Equals', Value 'Value > 27 Inch'. Below these is a '+ Add Expression' button. A 'Grouping Logic' section follows, with a text input containing '(1 or 2) and 3' and a 'PREFER' button. Another '+ Add Expression' button is present. A second 'Grouping Logic' section has a text input containing '1'. At the bottom, a 'Run-time Message' section contains the text '4K Display Preferred'.

14. Click Save in the top right corner of the header section of the editor.

15. The system will give a Save successful confirmation message if there are no validation errors and enable the Activate button..

16. Click the Activate button.

17. Note Activation successfully message is displayed if there are no errors and CML ready for use at runtime.

Preference Rule - It is NOT mandatory to satisfy the preference rule condition. The configurator tries to satisfy it but does not have to. The key difference here is that the user is allowed to override the preferred choice by the engine.



Salesforce Feature Name
Winter '26

Usage: Can be used to set default values or preferences that the user is allowed to change.

Eg: In the above use case, the user is allowed to change the Display even if the Display_Size is 27 Inch and Graphics is MSI Gaming GeForce RTX 3060.



Exclude Rule

Business Objective

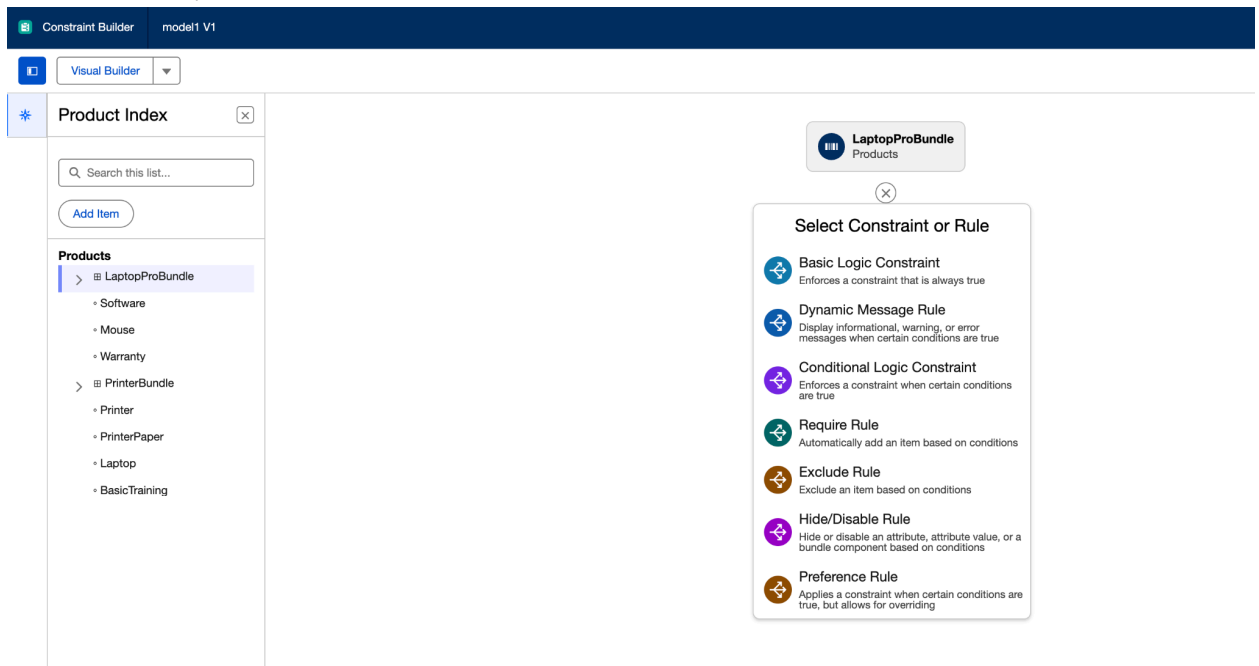
The primary function of the Exclude rule is dynamically enforcing the exclusion of specific components based on predefined conditions. This not only saves time and effort but also reduces the risk of errors..

Use Cases

This example uses the bundle product LaptopProBundle, which includes a PrinterBundle, which in turn includes a Printer and a PrinterPaper component. The "Exclude" rule ensures that when the sales representative selects a Printer with attribute printer as '3D' the PrinterPaper component is excluded from the bundle configuration.

Design Time Configuration

1. Open the LaptopProBundle constraint model in the Visual Builder.
2. Check if the model is Active and Deactivate it before proceeding to edit.
3. In the Visual Builder Left panel, select LaptopProBundle under the list of Products
4. In the main panel, click the '+' icon under the LaptopProBundle to view the available constraint templates.





5. Select Exclude Rule.
6. Write a constraint Summary : “Exclude Warranty”
7. In the top section of the constraint Card click ‘+ Add Expression’
8. Complete the expression definition as follows:
 - a. **LHS Element:** Mouse
 - b. **Operator:** == < Less than
 - c. **RHS Element:** Value: 1
9. In the bottom section (after exclude) of the constraint Card ‘Add Expression’ Complete the expression definition as follows:
 - a. Item: Warranty
10. Enter a run-time Message: Warranty excluded when no mouse selected.

The screenshot displays the Salesforce Visual Builder interface. On the left, the 'Product Index' sidebar shows a tree view of products under 'LaptopProBundle', including Software, Mouse, Warranty, Printer, PrinterPaper, Laptop, and BasicTraining. The main editor area shows a 'LaptopProBundle Products' card with a plus icon. Below it, a constraint card titled 'Exclude Warranty' is expanded. The card has a header 'Exclude Rule' and a description 'Exclude an item based on conditions.' The main body contains an expression builder with 'Mouse' as the LHS element, '< Less than' as the operator, and 'Value > 1' as the RHS element. Below the expression builder is a 'Grouping Logic' section with a dropdown set to '1'. At the bottom, there is an 'EXCLUDE' button, an 'Item' field set to 'Warranty', a 'Run-time Message' field with the text 'Warranty excluded when no mouse selected', and a 'Save' button in the top right corner.

11. Click Save in the top right corner of the header section of the editor.
12. The system will give a Save successful confirmation message if there are no validation errors and enable the Activate button..
13. Click the Activate button.
14. Note Activation successfully message is displayed if there are no errors and CML ready for use at runtime.



Hide Attribute Values Rule

Business Objective

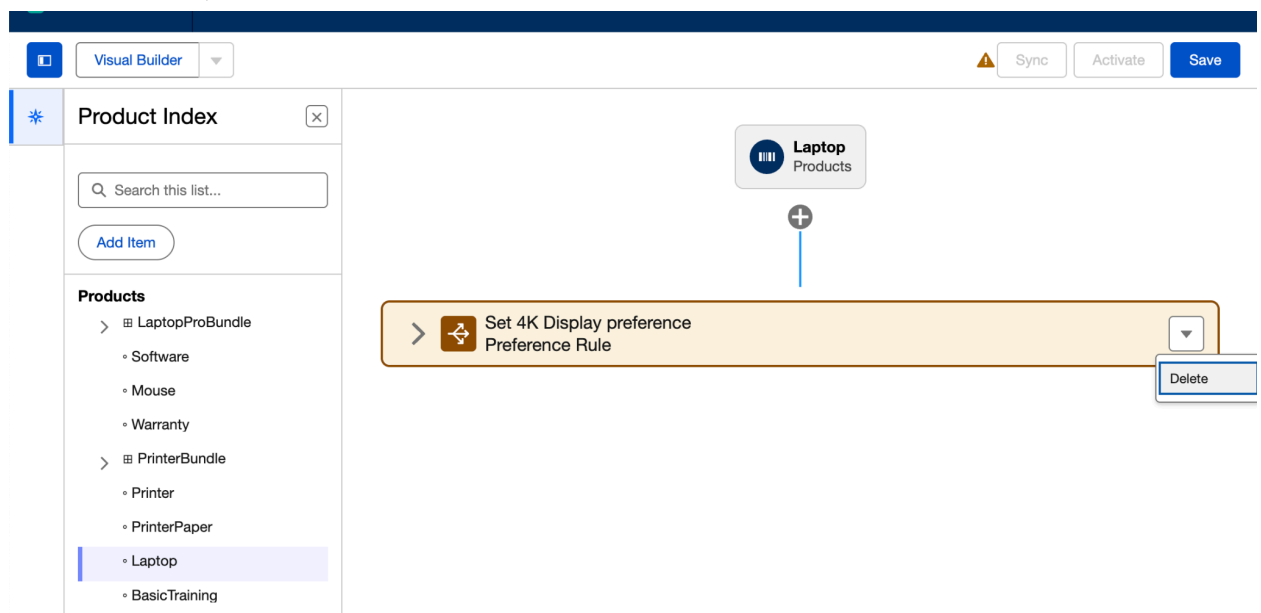
To hide attribute value(s).

Use Cases

When “27 Inch” is selected for Screen Size, hide all the attribute value options in Display_Resolution except “4k Built-in Display”.

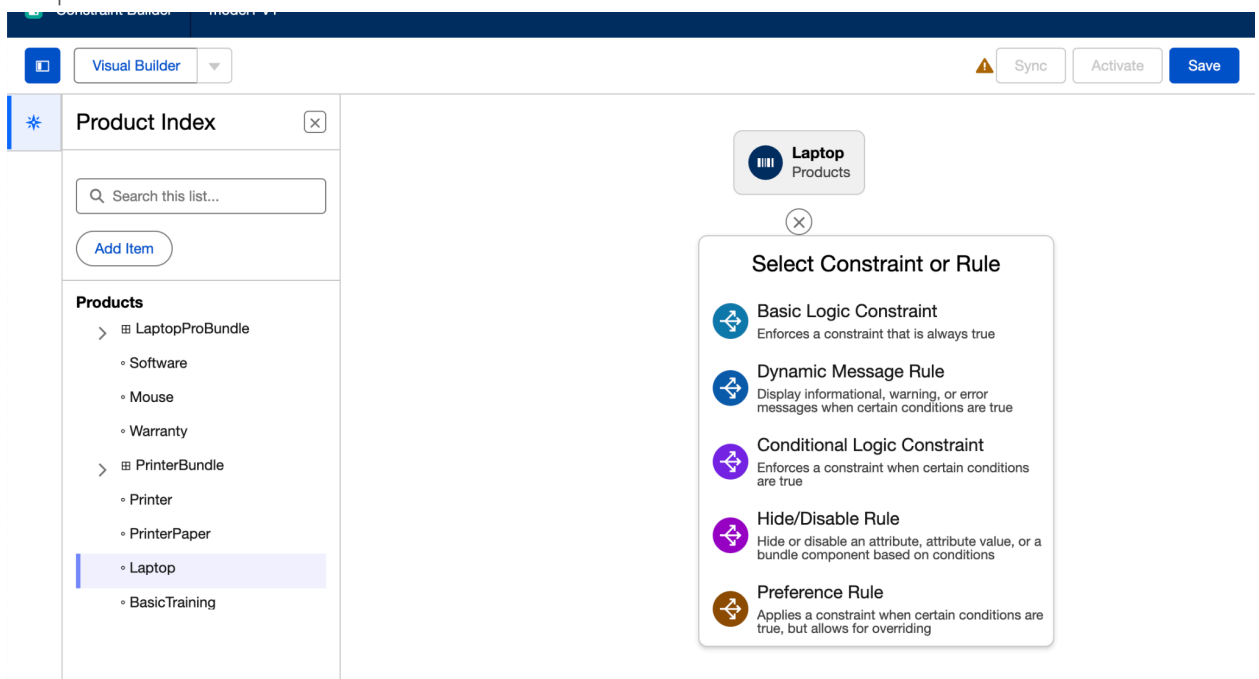
Design Time Configuration

1. Open the LaptopProBundle constraint model in the Visual Builder.
2. Check if the model is Active and Deactivate it before proceeding to edit.
3. In the Visual Builder Left panel, select Laptop under the list of Products
4. Delete the “Set 4K Display Preference” rule. (this step is not required, but to avoid any conflict in rules).





5. In the main panel, click the '+' icon under the Laptop to view the available constraint templates.



6. Select Hide/Disable Rule.
7. Write a constraint Summary : "Hide Display values"
8. In the top section of the constraint Card click '+ Add Expression'
9. Complete the expression definition as follows:
 - a. **LHS Element:** Laptop : Attribute: Display_Size
 - b. **Operator:** == Equals
 - c. **RHS Element:** Value: 27 Inch
10. Review that "Hide" is selected by default. Do not change.



Visual Builder

Sync Activate Save

Product Index

Search this list...

Add Item

Products

- LaptopProBundle
 - Software
 - Mouse
 - Warranty
- PrinterBundle
 - Printer
 - PrinterPaper
 - Laptop
 - BasicTraining

Laptop Products

Hide Display values

Hide/Disable Rule

Hide or disable an attribute, attribute value, or a bundle component based on conditions.

Element Operator Element

1 Laptop ▶ Display_Size == Equ... Value ▶ 27 Inch

Add Expression

Grouping Logic

1

Group expressions within '()', and connect the groups using 'and', 'or', or 'xor' logic. When the logic contains mixed operators, always group expressions by using '()' to define the exact evaluation order.

Hide

Hide

Disable

11. In the bottom section (after Hide) of the constraint Complete the definition as follows:

- Element: Attribute Value : Display :
- Value: 1080p Built-in Display, 2k Built-in Display



The screenshot shows the Salesforce Visual Builder interface. On the left, the 'Product Index' sidebar lists products under 'LaptopProBundle' and 'PrinterBundle'. The 'Laptop' product is selected. The main workspace shows a 'Hide Display values' rule configuration. The rule is titled 'Hide Display values' and is a 'Hide/Disable Rule'. It contains one expression: 'Laptop ► Display_Size == Equ... Value ► 27 Inch'. The 'Grouping Logic' is set to '1'. The 'Hide' dropdown is selected. The 'Element' section shows 'Attribute Value ► Display ► 1080p Built-in Display' and '2k Built-in Display'.

12. Click Save in the top right corner of the header section of the editor.
13. The system will give a Save successful confirmation message if there are no validation errors and enable the Activate button..
14. Click the Activate button.
15. Note Activation successfully message is displayed if there are no errors and CML ready for use at runtime.

Run Time Behavior

As a sales representative, on configuring the LaptopproBundle, if I choose value “27 Inch” for the Display_Size attribute of the Laptop, then attribute value options of the Display attribute will be hidden except “4k Built-in Display”. Also notice, the selected attribute value won't be hidden. If “4k Built-in Display” is selected, then it will be the only available attribute value. If “2k Built-in Display” is selected, then both “2k Built-in Display” and “4k Built-in Display” will be displayed.



Disable Components Rule

Business Objective

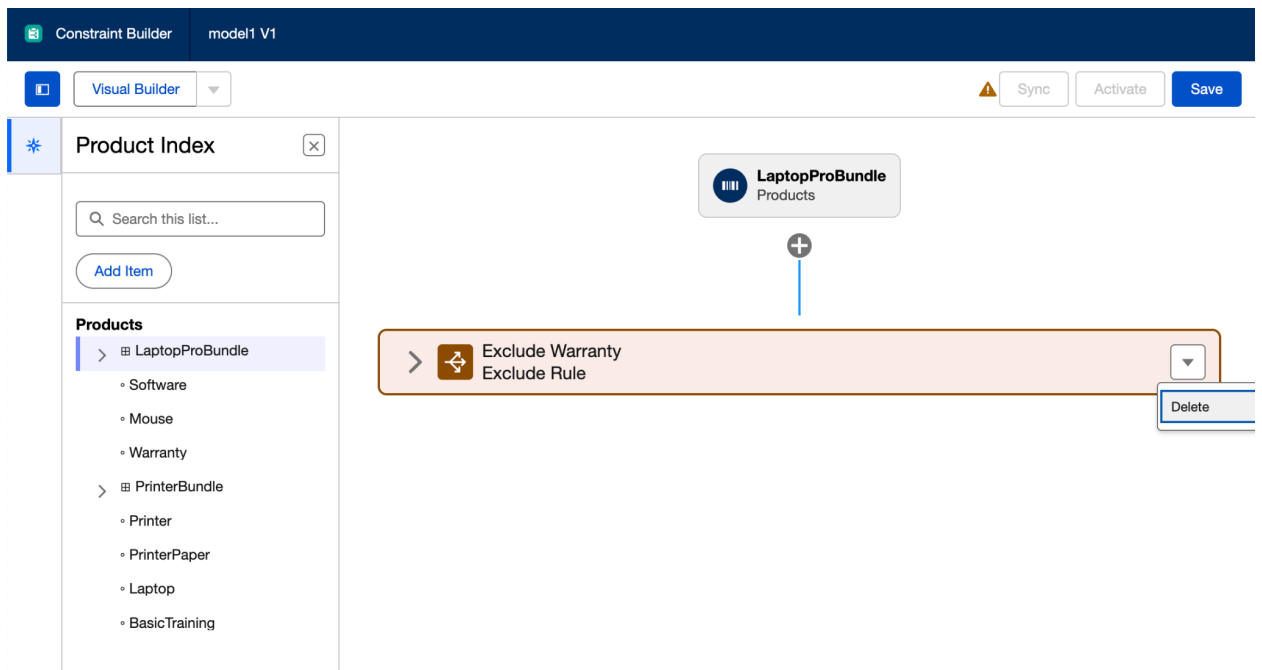
To disable child components..

Use Cases

If Laptop is selected in Laptop Pro Bundle, disable any changes to Warranty.

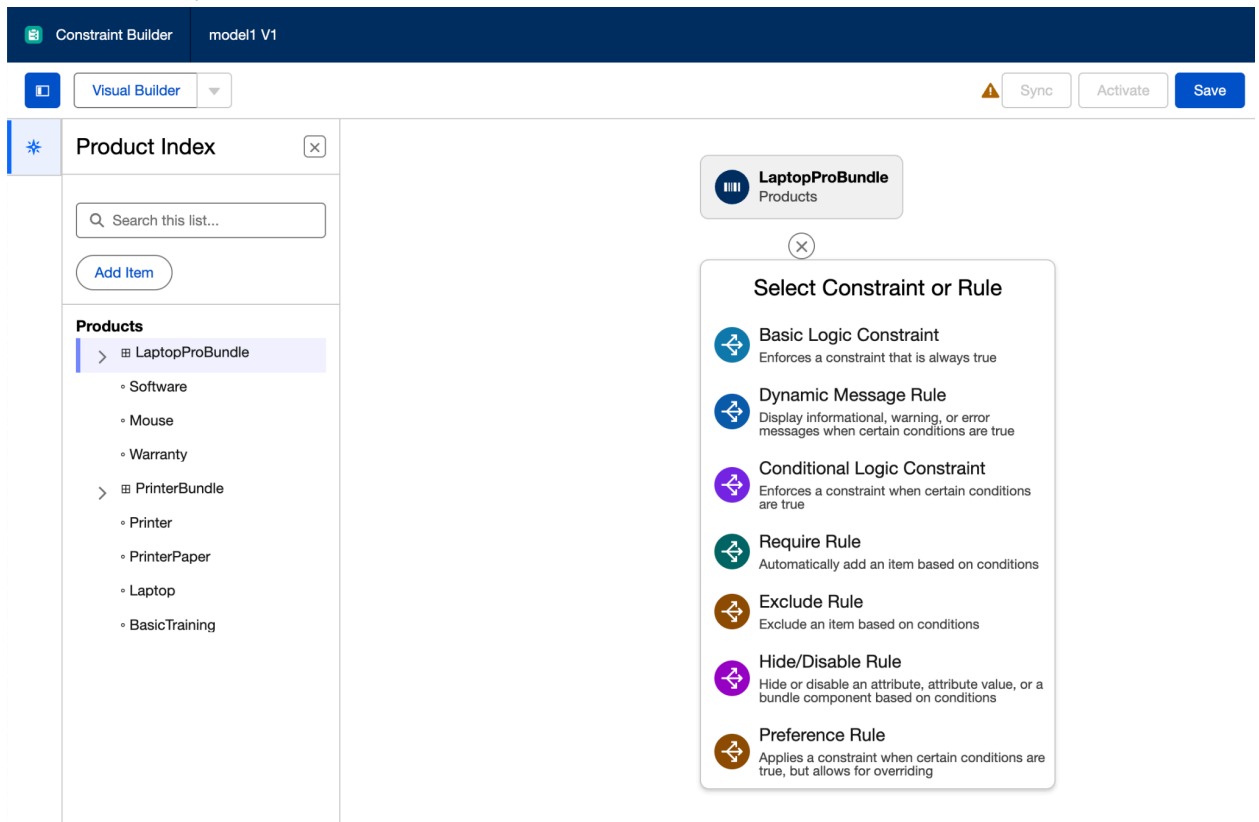
Design Time Configuration

1. Open the LaptopProBundle constraint model in the Visual Builder.
2. Check if the model is Active and Deactivate it before proceeding to edit.
3. In the Visual Builder Left panel, select LaptopProBundle under the list of Products
4. Delete the “Exclude Warranty” rule. (this step is not required, but to avoid any conflict in rules).





5. In the main panel, click the '+' icon under the LaptopProBundle to view the available constraint templates.



6. Select Hide/Disable Rule.
7. Write a constraint Summary : “Disable Warranty”
8. In the top section of the constraint Card click ‘+ Add Expression’
9. Complete the expression definition as follows:
 - a. **LHS Element:** Mouse
 - b. **Operator:** == < Less than
 - c. **RHS Element:** Value: 1
10. Review that the rule action “Hide” is selected by default. Change to “Disable”.



Visual Builder

Sync Activate Save

Product Index

Search this list...

Add Item

Products

- LaptopProBundle
 - Software
 - Mouse
 - Warranty
- PrinterBundle
 - Printer
 - PrinterPaper
 - Laptop
 - BasicTraining

LaptopProBundle Products

+

Disable Warranty

Hide/Disable Rule

Hide or disable an attribute, attribute value, or a bundle component based on conditions.

1 Element Mouse Operator < Less ... Element Value 1

+ Add Expression

Grouping Logic 1

1

Group expressions within '()', and connect the groups using 'and', 'or', or 'xor' logic. When the logic contains mixed operators, always group expressions by using '()' to define the exact evaluation order.

Disable

Hide

✓ Disable

11. In the bottom section (after Disable) of the constraint Complete the definition as follows:

- Element: Product : Warranty



The screenshot displays the Salesforce Visual Builder interface. On the left, a sidebar titled 'Product Index' contains a search bar and a list of products under the 'Products' section: LaptopProBundle (selected), Software, Mouse, Warranty, PrinterBundle, Printer, PrinterPaper, Laptop, and BasicTraining. The main workspace shows a configuration for the 'LaptopProBundle' with a 'Disable Warranty' rule. The rule is titled 'Disable Warranty' and is a 'Hide/Disable Rule'. It contains a single condition: 'Mouse' (Element) is '< Less ...' (Operator) 'Value' (Element) '1'. The rule is set to 'Disable' the 'Warranty' element. The 'Save' button is highlighted in blue in the top right corner.

12. Click Save in the top right corner of the header section of the editor.
13. The system will give a Save successful confirmation message if there are no validation errors and enable the Activate button..
14. Click the Activate button.
15. Note Activation successfully message is displayed if there are no errors and CML ready for use at runtime.

Run Time Behavior

As a sales representative, on configuring the LaptopproBundle, if Mouse is not selected, then Warranty will be disabled. If Warranty is selected, then it can't be de-selected and the quantity can't be changed. If Warranty is not selected, then it can be selected.



Non Blocking UI

Business Objective

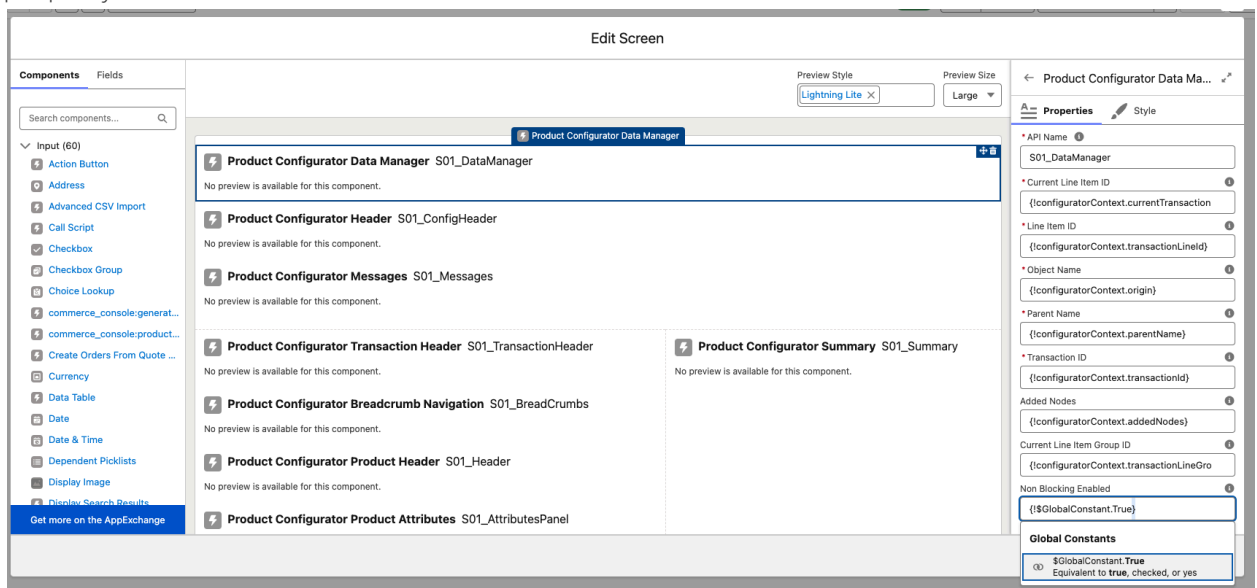
Ability to not block the sales rep at the time of configuration each time they change an attribute

Use Cases

If attributes of Laptop are changed ,the user no longer gets a spinner but instead can continue selecting other attributes.

Design Time Configuration

1. In app launcher search for 'Product Configuration Flows' and check what flow is being used.
2. From setup in the quick find box search for 'Flows' and click on the flow that's being used (from step 1)
3. On the flow builder screen click on ProductConfigurationUI
4. Click on the Product Configuration Data Manager and set Non Blocking Enabled property to true



5. If you are using default flow (from step 1), this is all you need to do. Click Done -> Save As New Version -> Activate. Non blocking should be enabled now.



6. If you are not using the default flow, pass down non blocking enabled to all other screen components in the flow
7. Click on all the screen components in the flow and look for non blocking enabled property on them.
8. To set the property - You'll see S01_DataManager screen component when you click on Non Blocking Enabled text box, select that and from there select the isNonBlockingEnabled property
9. Do this for all the screen components that have the non blocking property on them
10. This will make sure all the components have the correct non blocking value passed on to them from data manager
11. Click Done -> Save As New Version -> Activate. Non blocking should be enabled now.

Run Time Behavior

As a sales representative, on configuring the LaptopproBundle, on selecting attributes , users no longer have to wait for the engine to complete processing and be blocked before they can make their next selection. Non Blocking UI for attributes enables users to make various attribute selections for a given component without being blocked.